

# Jiaju Wu

Mathematics undergraduate, Peking University

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## RESEARCH

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Transformer computation, mechanistic interpretability, and algorithmic generalization.

### Shared Weights, Selected Computations: How Looped Transformers Route What Each Loop Does

Jiaju Wu, Yi Hu, Muhan Zhang. 2026. arXiv:2609.39892. Submitted to ICLR 2027.

Graph-walk experiments test how the entering hidden state selects different transitions in a frozen looped Transformer. Attention interventions examine the route of this control. Results are specific to the studied tasks and models.

Paper | Code, archived results, and reproduction guide

## EDUCATION

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### Peking University, School of Mathematical Sciences | Aug 2024 - present

Undergraduate. GPA: 3.79 / 4.00.

Selected coursework: Mathematical Analysis, Advanced Algebra, Probability Theory, Mathematical Statistics, Data Structures and Algorithms, Deep Learning Theory.

## SELECTED PROJECTS

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### Learning Dynamics and Schedule Transfer | Course project

Source-only calibration of a residual correction from cosine to WSD learning-rate schedules. The reported correction improves all 15 same-scale targets relative to MPL; evaluation uses existing public curves. Code and results.

### Grokking on Small Algorithmic Datasets | Team course project

With Kairui Li and Kehan Huang: a training and evaluation framework for delayed generalization on modular arithmetic and group operations. Code and report.

### SMS Student Union Academic Department Website | Campus software

React and TypeScript website for mathematical problems, resources, and events. Markdown/KaTeX rendering and configuration-based content maintenance. Repository.

## TEACHING AND SERVICE

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### Teaching Assistant, Python and AI Foundations | Summer 2025

Supported programming and Avalon-agent projects; maintained the Tuvalon teaching platform.

### Head of Academic Department, SMS Student Union | Mar 2025 - Mar 2026

Organized academic activities and mock exams; launched the department website.

## SELECTED HONORS

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Chinese Mathematical Olympiad (CMO), Gold Medal, 2023.

Lingjun Linghang University Scholarship, Peking University, 2025.

## TOOLS

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Python, PyTorch, NumPy, scikit-learn, TypeScript, React, Git, LaTeX.